BLM3620 Digital Signal Processing

Dr. Ali Can KARACA

ackaraca@yildiz.edu.tr

Yıldız Technical University – Computer Engineering

Lecture #14 – Filter Design Basics

- Relation of z-Transform and DTFT
- Examples
- Filter Design Using z-Plane
- MATLAB Example

Course Materials

Important Materials:

- James H. McClellan, R. W. Schafer, M. A. Yoder, *DSP First Second Edition*, Pearson, 2015.

- Lizhe Tan, Jean Jiang, *Digital Signal Processing: Fundamentals and Applications*, Third Edition, Academic Press, 2019.

Auxilary Materials:

- Prof. Sarp Ertürk, *Sayısal İşaret İşleme*, Birsen Yayınevi.

- Prof. Nizamettin Aydin, DSP Lecture Notes.

- J. G. Proakis, D. K. Manolakis, *Digital Signal Processing Fourth Edition*, Peason, 2014.

- J. K. Perin, *Digital Signal Processing*, *Lecture Notes,* Standford University, 2018.

Syllabus

Week	Lectures
1	Introduction to DSP and MATLAB
2	Sinuzoids and Complex Exponentials
3	Spectrum Representation
4	Sampling and Aliasing
5	Discrete Time Signal Properties and Convolution
6	Convolution and FIR Filters
7	Frequency Response of FIR Filters
8	Midterm Exam
9	Discrete Time Fourier Transform and Properties
10	Discrete Fourier Transform and Properties
11	Fast Fourier Transform and Windowing
12	-
13	z- Transforms
14	FIR Filter Design and Applications

Vize ve öncesi 1 soru

TF olarak 4 soru toplam

z dönüşümü

Fourier dönüşümü

DFT - FFT

For more details -> Bologna page: http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=5730&aid=3

Z-Transform EXAMPLE

- ANY SIGNAL has a z-Transform:

Transfer Function

$$X(z) = \sum_n x[n] z^{-n}$$

$$H(z) = \sum_n h[n] z^{-n}$$

Example 7.1

n	$n < -1$	-1	0	1	2	3	4	5	$n > 5$
$x[n]$	0	0	2	4	6	4	2	0	0

$$X(z) = ?$$

$$X(z) = 2 + 4z^{-1} + 6z^{-2} + 4z^{-3} + 2z^{-4}$$

© 2003-2016, JH McClellan & RW Schafer

FREQUENCY RESPONSE ?

- Same Form:

$\hat{\omega}$ – Domain

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} b_k e^{-j\hat{\omega}k}$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} b_k (e^{j\hat{\omega}})^{-k}$$

$$z = e^{j\hat{\omega}}$$

z – Domain

$$H(z) = \sum_{k=0}^{M} b_k z^{-k}$$

SAME COEFFICIENTS

If we have single complex exponential signal....

SINUSOIDAL RESPONSE

- $x[n]$ = SINUSOID => $y[n]$ is SINUSOID
- Get MAGNITUDE & PHASE from $H(z)$

$$\boxed{\begin{aligned}&\text{if}\quad x[n] = e^{j\hat{\omega}n}\\ &\text{then}\quad y[n] = H(e^{j\hat{\omega}})e^{j\hat{\omega}n}\\ &\text{where}\quad H(e^{j\hat{\omega}}) = H(z)\big|_{z=e^{j\hat{\omega}}}\end{aligned}}$$

Exercise-1

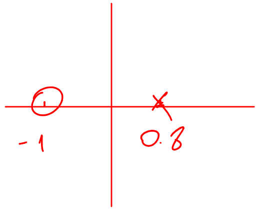
POP QUIZ

- Given: $\boxed{H(z) = \dfrac{2 + 2z^{-1}}{1 - 0.8z^{-1}}}$

$\rightarrow 2z+2 \rightarrow z=-1$ $\rightarrow$ zeros

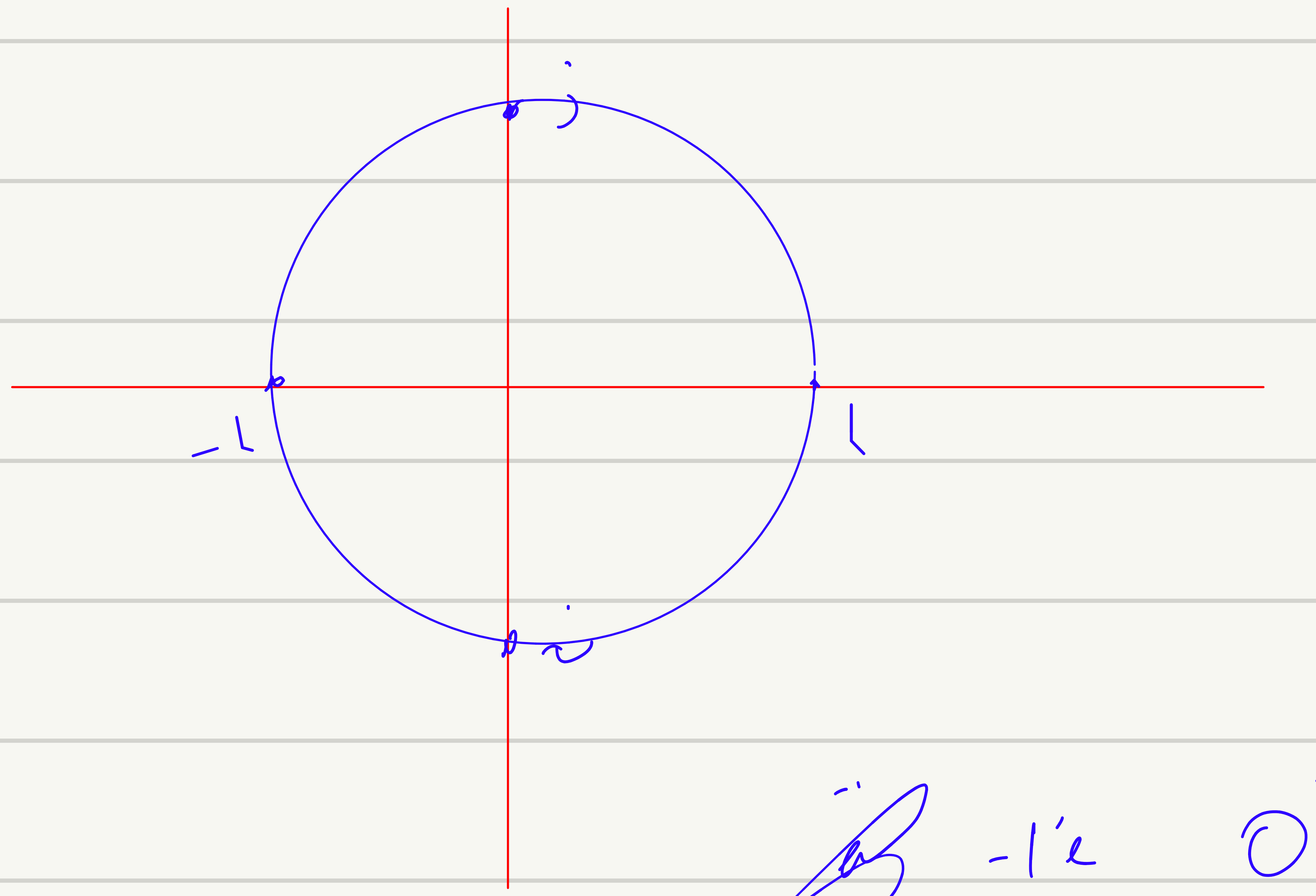
$z - 0.8$ $\rightarrow$ poles 0.8

- Find the Impulse Response, $h[n]$



- Find the output, $y[n]$
  - When

$$\boxed{x[n] = \cos(0.25\pi n)}$$



$z = e^{+j\hat{\omega}}$

$\hat{\omega} = 0 \qquad z = 1$

$\hat{\omega} = \frac{\pi}{2} \qquad z = j$

$\hat{\omega} = \pi \qquad z = -1$

$\hat{\omega} = \frac{3\pi}{2} \qquad z = -j$

-1'e O → zero eklesek π noktasını bastıran filtremiz olmuş olur

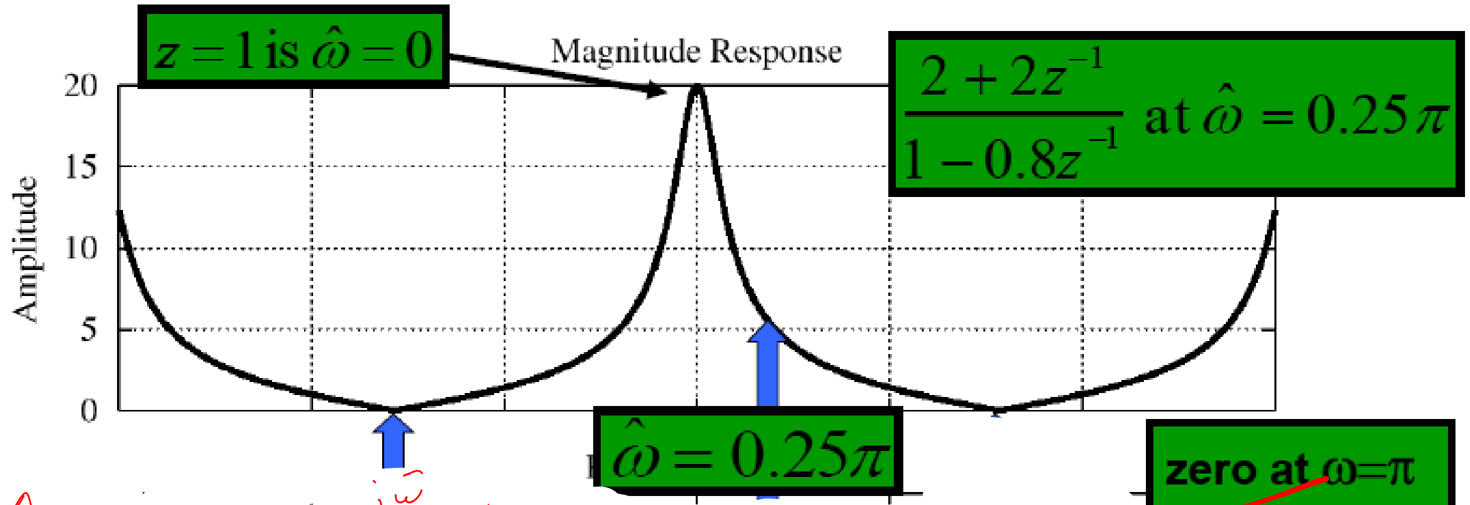
-1'e X eklesek " " geçiren " " " .

X → pole

-1'e yakın yere eklesen de etki eder 0.8

Exercise-1

Evaluate FREQ. RESPONSE



$\hat{w} = 0 \rightarrow z = e^{j\hat{w}} = 1$

$H(z) = \frac{2z+2}{z-0.8}\Big|_{z=1} \rightarrow 20$

$\hat{w} = \pi/2 \rightarrow z = j$

$H(z) = \frac{2j+2}{j-0.8}\Big|_{z=j} \rightarrow \frac{2\sqrt{2}}{\sqrt{1+0.64}} \approx 2,2$

Exercise-1

POP QUIZ: Eval Freq. Resp.

- Given: $$H(z) = \frac{2+2z^{-1}}{1-0.8z^{-1}}$$

cos(πn) olsaydı

$z = e^{j\hat{\omega}} = -1$

$\frac{-2+2}{2-0.8} \rightarrow 0$

- Find output, $y[n]$, when
  - Evaluate at

$$x[n] = \cos(0.25\pi n)$$

$\hat{\omega} = 0.25\pi$

$$z = e^{j0.25\pi} \rightarrow \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}$$

$$H(z) = \frac{2+2(\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2})}{1-0.8e^{-j0.25\pi}} = 5.182e^{-j1.309}$$

$$y[n] = \underbrace{5.182}_{\text{genlik}}\cos(0.25\pi n - \underbrace{0.417\pi}_{\text{faz}})$$

$H(z) = 1 - z^{-1} \rightarrow \frac{z-1}{z}$ → zeros → 1

z → pole = 0

* $\cos(\pi n) \rightarrow \hat{w} = \pi \quad z = -1$

$\frac{-2}{-1} = 2 \quad \arctan(0) = 0$

$2e^{j0} \rightarrow \cos(\pi n) \cdot 2$

* $\cos(0.5\pi n) \rightarrow \hat{w} = 0.5\pi \quad z = j$

$\frac{j-1}{j} \rightarrow -1-j \rightarrow \sqrt{2} \cdot e^{+j\pi/4} \rightarrow \sqrt{2} \cdot \cos(0.5\pi n + \pi/4) = y[n]$

→ $j-1$ in açısı $\arctan\left(\frac{1}{-1}\right) \rightarrow \frac{3\pi}{4}$

→ $j = \frac{\pi}{2}$

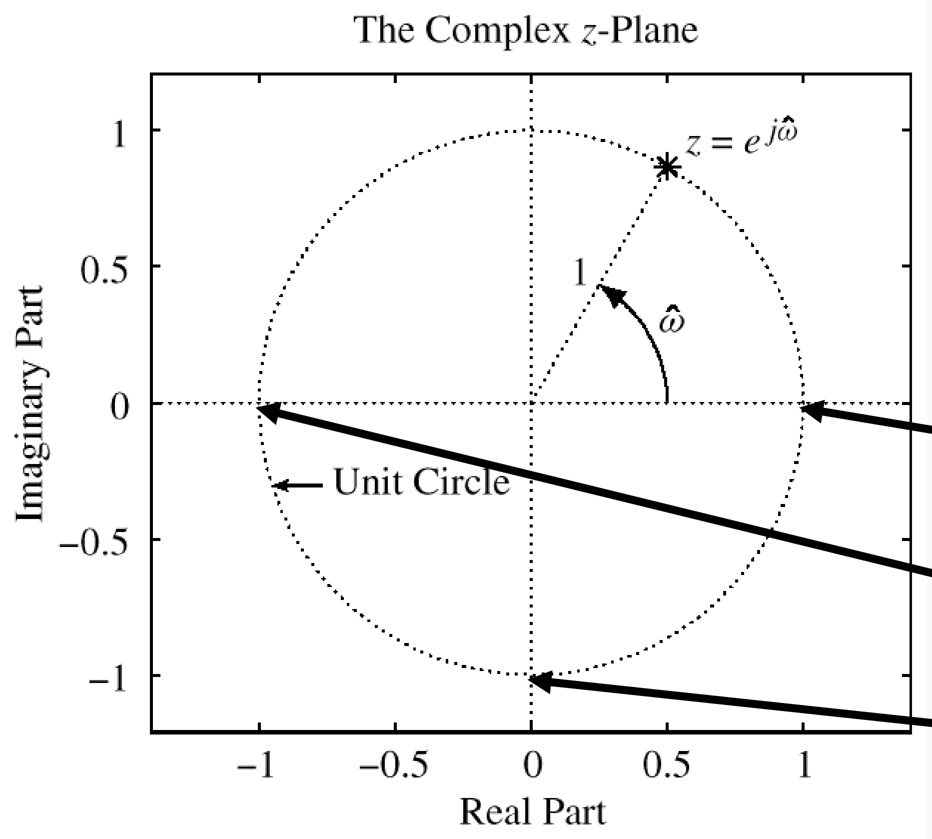
pay − payda → $\frac{3\pi}{4} - \frac{\pi}{2} = \frac{\pi}{4}$

* $\cos(\pi n) + \cos(0.5\pi n)$ → ikisinin ayrı ayrı hesaplanıp toplanmış hali.

UNIT CIRCLE: RECAP

- MAPPING BETWEEN

$$z \text{ and } \hat{\omega}$$



$$\boxed{z = e^{j\hat{\omega}}}$$

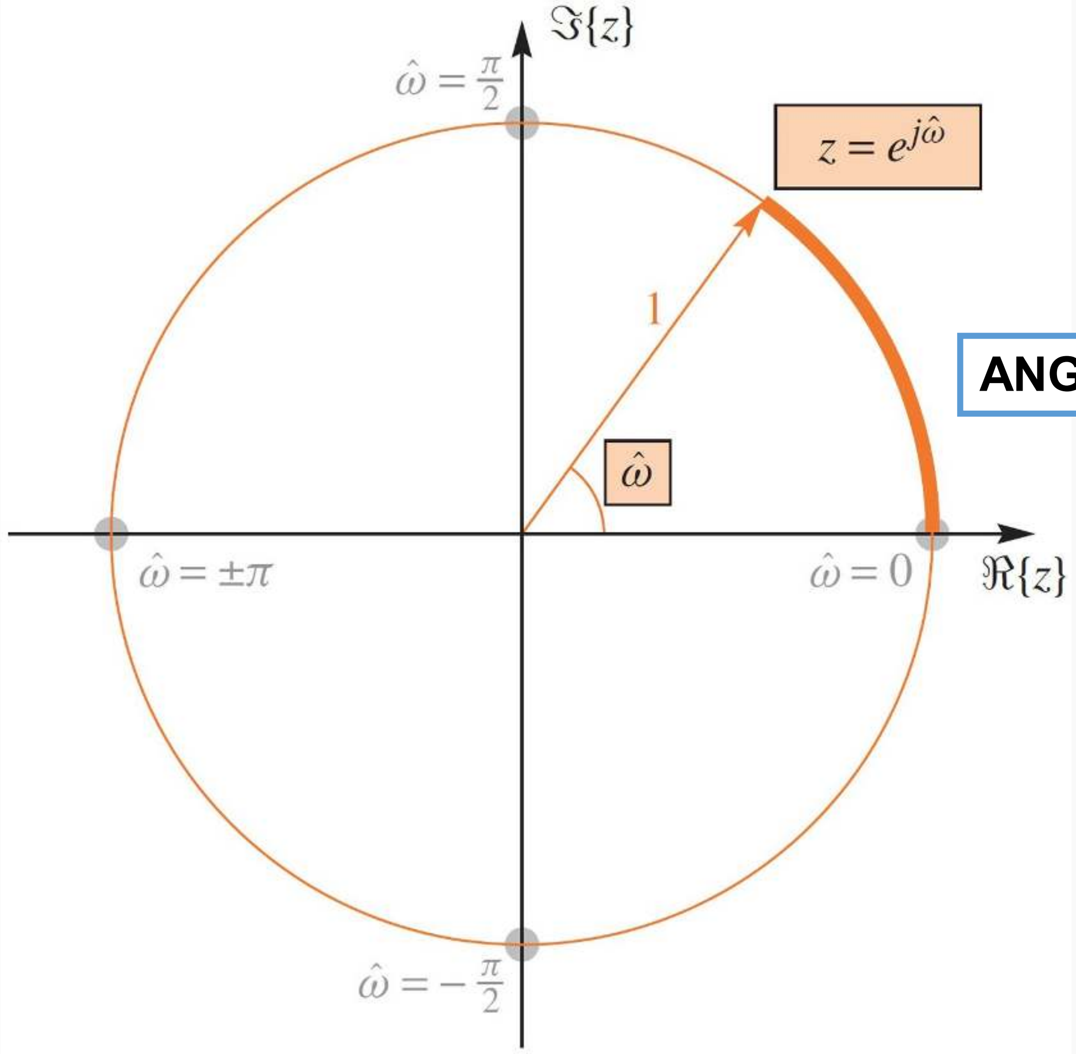
$$z = 1 \quad \leftrightarrow \quad \hat{\omega} = 0$$

$$z = -1 \quad \leftrightarrow \quad \hat{\omega} = \pm\pi$$

$$z = \pm j \quad \leftrightarrow \quad \hat{\omega} = \pm\tfrac{1}{2}\pi$$

© 2003-2016, JH McClellan & RW Schafer

$$\boxed{H(e^{j\hat{\omega}}) = H(z)\Big|_{z=e^{j\hat{\omega}}}}$$



ANGLE is FREQUENCY

Frequency Response from poles and zeros

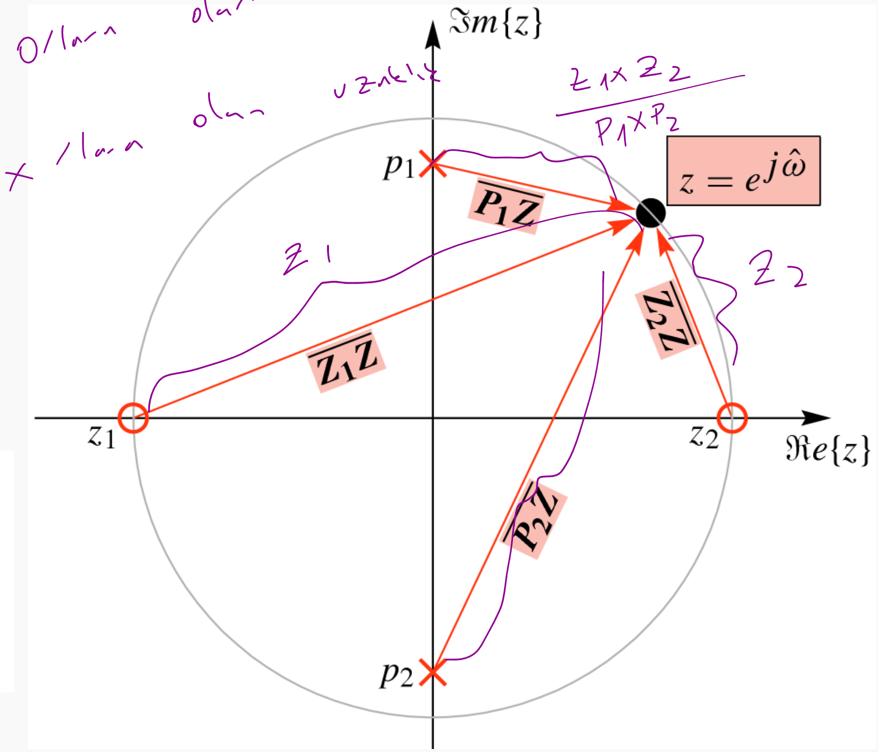
not: Origine koyulan yapılar sonucu çok etkilemez

$$|H(e^{j\hat{\omega}})| = G\,\frac{|e^{j\hat{\omega}} - z_1|\,|e^{j\hat{\omega}} - z_2|}{|e^{j\hat{\omega}} - p_1|\,|e^{j\hat{\omega}} - p_2|}$$

$$|H(e^{j\hat{\omega}})| = G\,\frac{\overline{Z_1Z}\cdot\overline{Z_2Z}}{\overline{P_1Z}\cdot\overline{P_2Z}}$$

→ 0'lara olan uzaklık

→ x'lere olan uzaklık



$$H(z) = G\,\frac{(z - z_1)(z - z_2)}{(z - p_1)(z - p_2)}$$

© 2003-2016, JH McClellan & RW Schafer

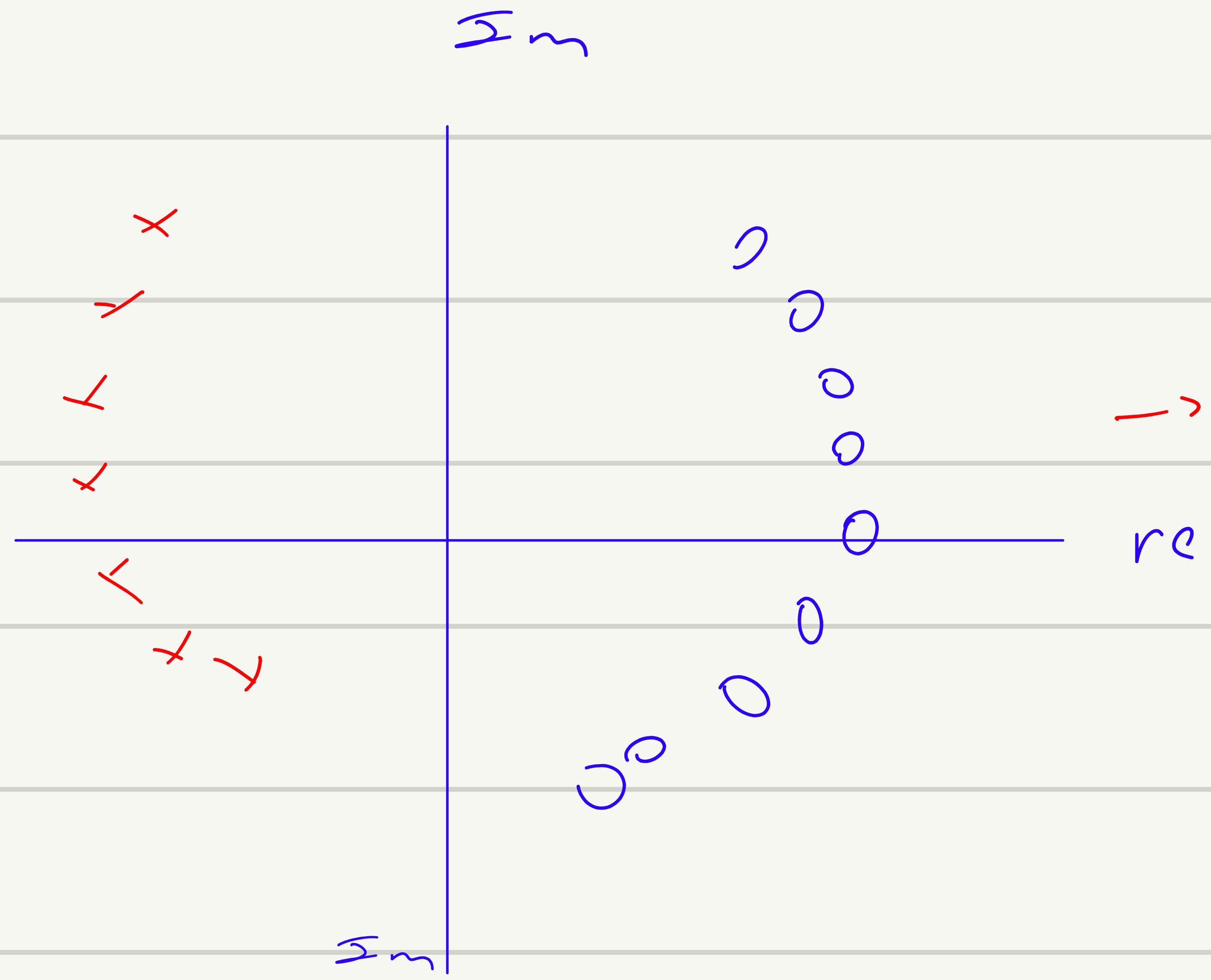
$$H(z) = \frac{z}{z - 0.8}$$

$\hat{\omega} = \pi/2 \quad z = j$

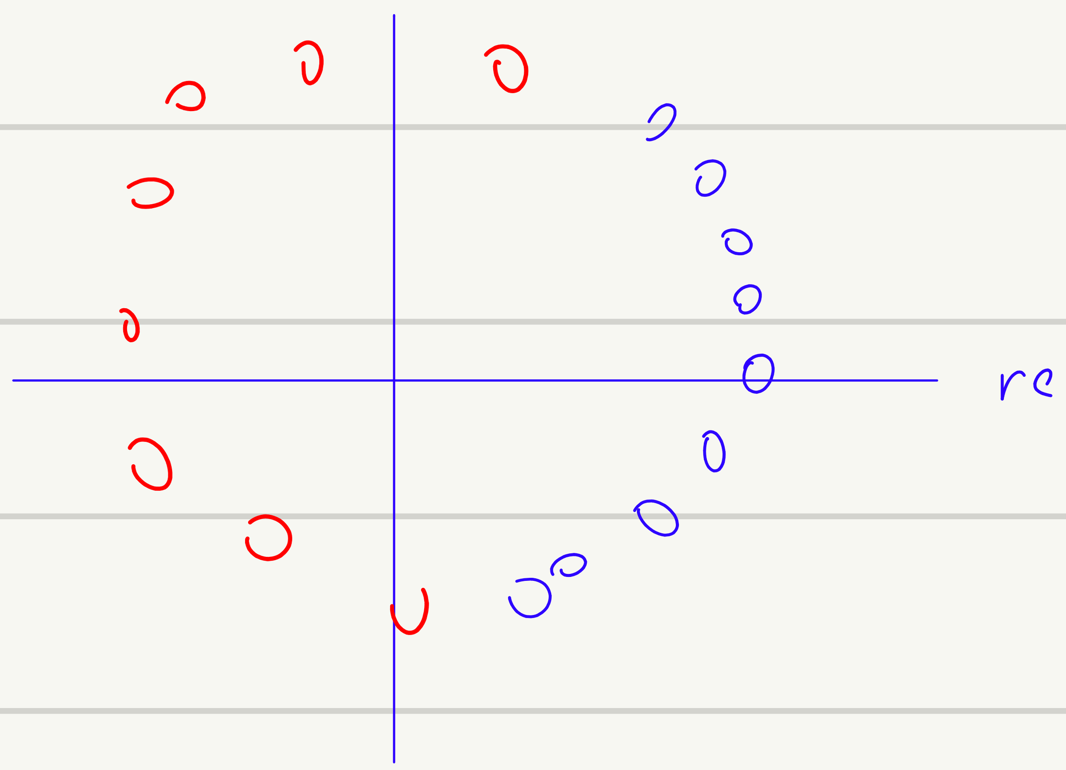
$$\frac{j}{j - 0.8} \to \frac{1}{\sqrt{1{,}64}} \to \frac{1}{\sqrt{1{,}64}} = 0.78$$

$\hat{\omega} = \pi \quad z = -1$

$$\frac{\sqrt{(-1)^2}}{\sqrt{(-1-0.8)^2}} \to \frac{1}{1.8}$$

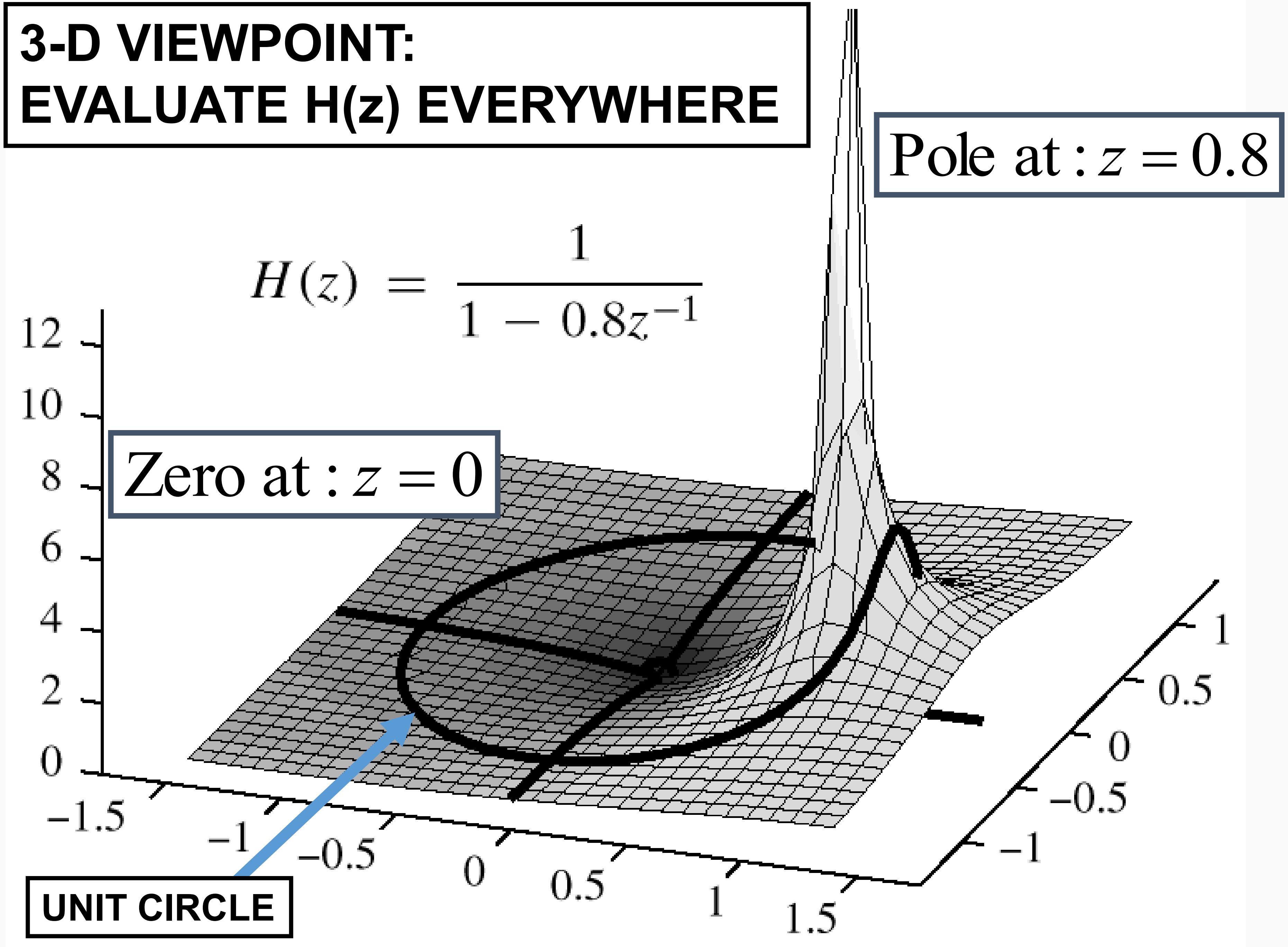


→ yüksekleri daha keskin geçiriyor

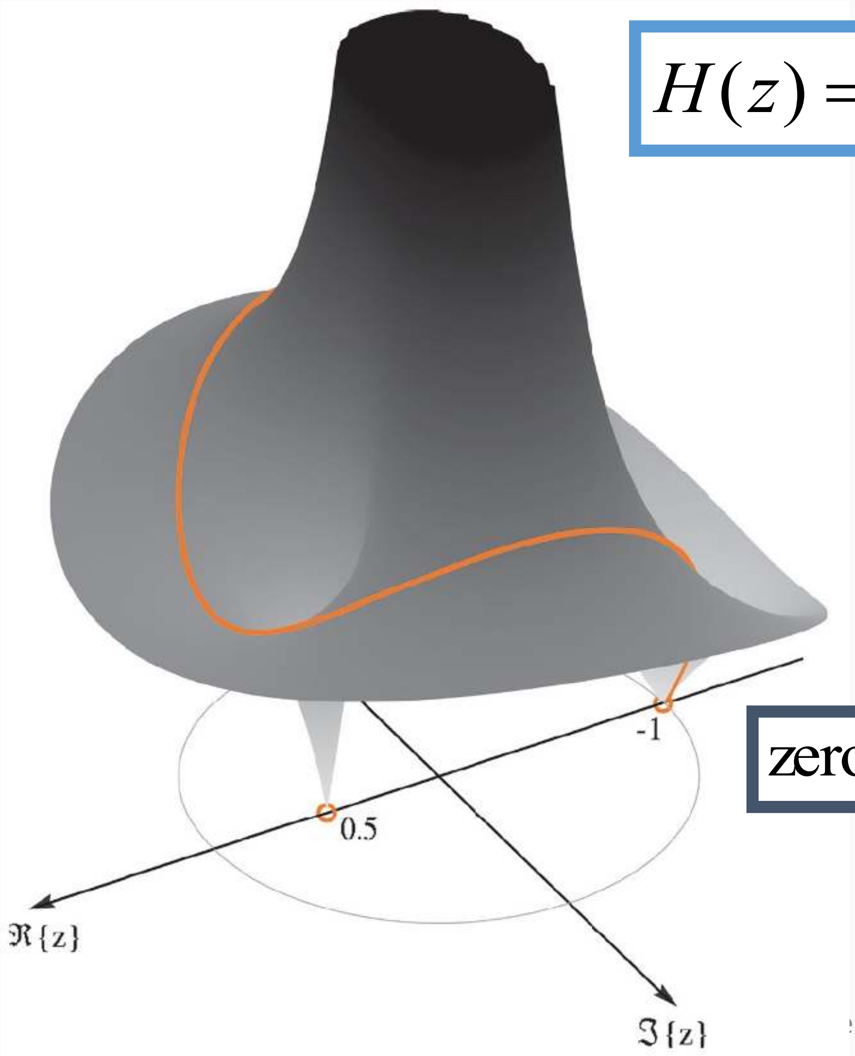


→ Bant geçiren

3-D VIEWPOINT:
EVALUATE H(z) EVERYWHERE



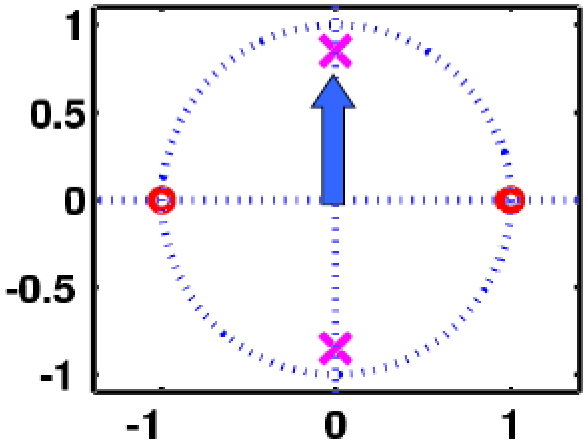
Evaluate H(z) on Unit Circle



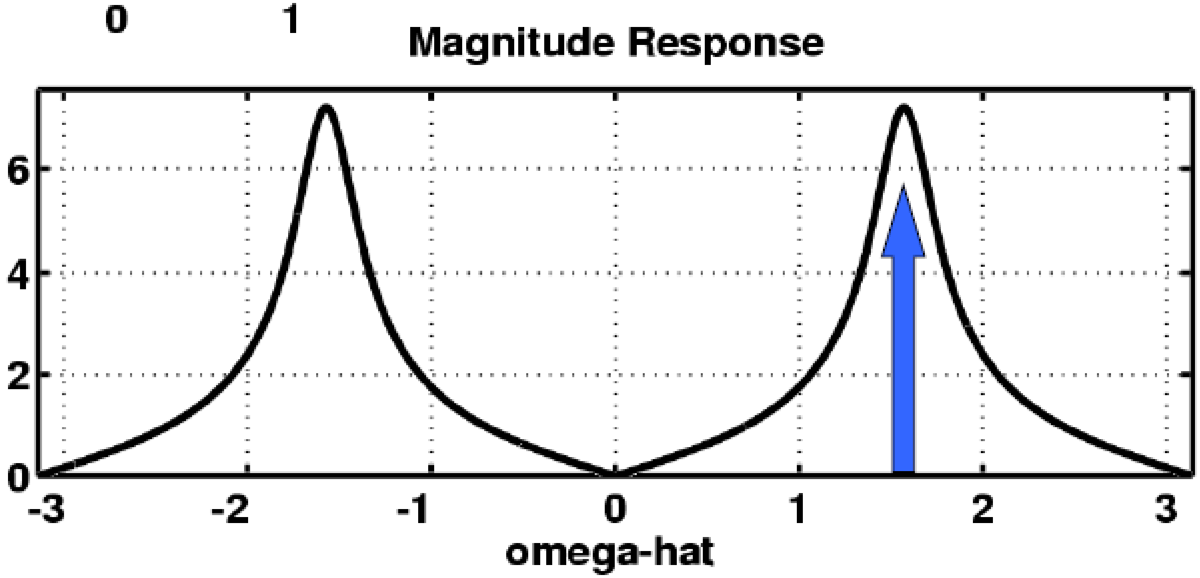
$$H(z) = (1 + z^{-1})(1 - 0.5z^{-1})$$

zeros at $z = -1$ and $z = 0.5$

FREQUENCY RESPONSE from POLE-ZERO PLOT



$$H(e^{j\hat{\omega}}) = \frac{1 - e^{-j2\hat{\omega}}}{1 + 0.7225e^{-j2\hat{\omega}}}$$



ZEROS of H(z) – example 2

- Find z, where H(z)=0
  - Interesting when z is ON the unit circle.

$$H(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}$$

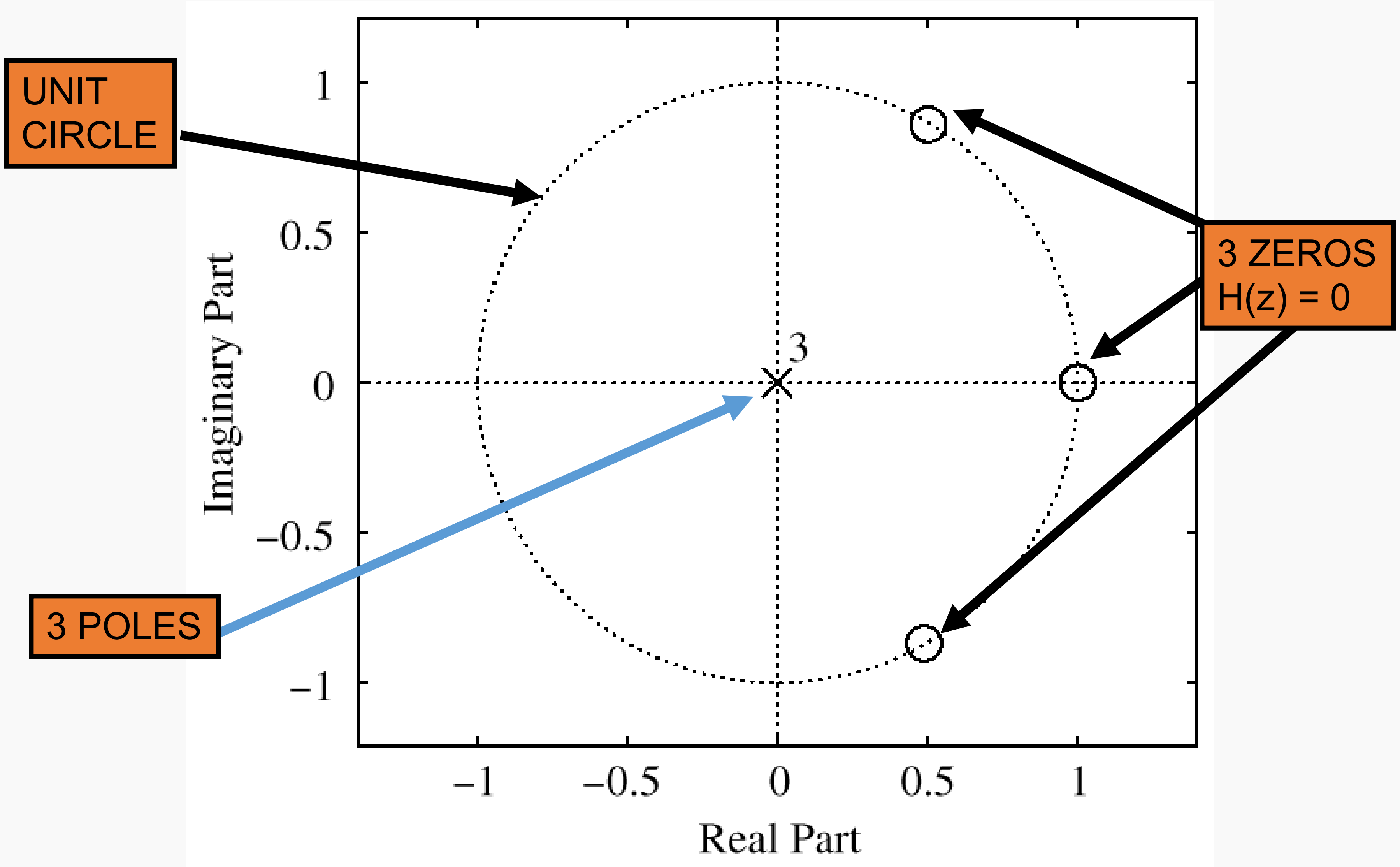
$\rightarrow \dfrac{z^3 - 2z^2 + 2z - 1}{z^3}$

$$H(z) = (1 - z^{-1})(1 - z^{-1} + z^{-2})$$

$$\boxed{\text{Roots}: z = 1, \tfrac{1}{2} \pm j\tfrac{\sqrt{3}}{2}} \qquad \boxed{e^{\pm j\pi/3}}$$

Recall: Roots occur in Conjugate pairs when coefficients are real

PLOT ZEROS in z-DOMAIN



POLES of H(z)

- Find z, where
  - FIR only has poles at z=0

$$H(z) \to \infty$$

$$H(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}$$

$$\boxed{H(z) = \frac{z^3 - 2z^2 + 2z \;\; - 1}{z^3}}$$

$$\boxed{\text{Three Poles at}: z = 0}$$

NULLING PROPERTY of H(z)

- When H(z)=0 on the unit circle.
  - Find inputs x[n] that give zero output

$$H(z) = 1 - 2z^{-1} + 2z^{-2} - z^{-3}$$

$$H(e^{j\hat{\omega}}) = 1 - 2e^{-j\hat{\omega}} + 2e^{-j2\hat{\omega}} - e^{-j3\hat{\omega}}$$

$$x[n] \longrightarrow \boxed{H(z)} \longrightarrow y[n]$$

$$\boxed{H(e^{j\pi/3}) = ?}$$

$$x[n] = e^{j(\pi/3)n}$$

$$y[n] = H(e^{j(\pi/3)}) \cdot e^{j(\pi/3)n}$$

© 2003-2016, JH McClellan & RW Schafer

NULLING PROPERTY of H(z)

- Evaluate H(z) at the input “frequency”

$$H(e^{j\hat{\omega}}) = 1 - 2e^{-j\hat{\omega}} + 2e^{-j2\hat{\omega}} - e^{-j3\hat{\omega}}$$

$$y[n] = H(e^{j\pi/3}) \cdot e^{j(\pi/3)n}$$

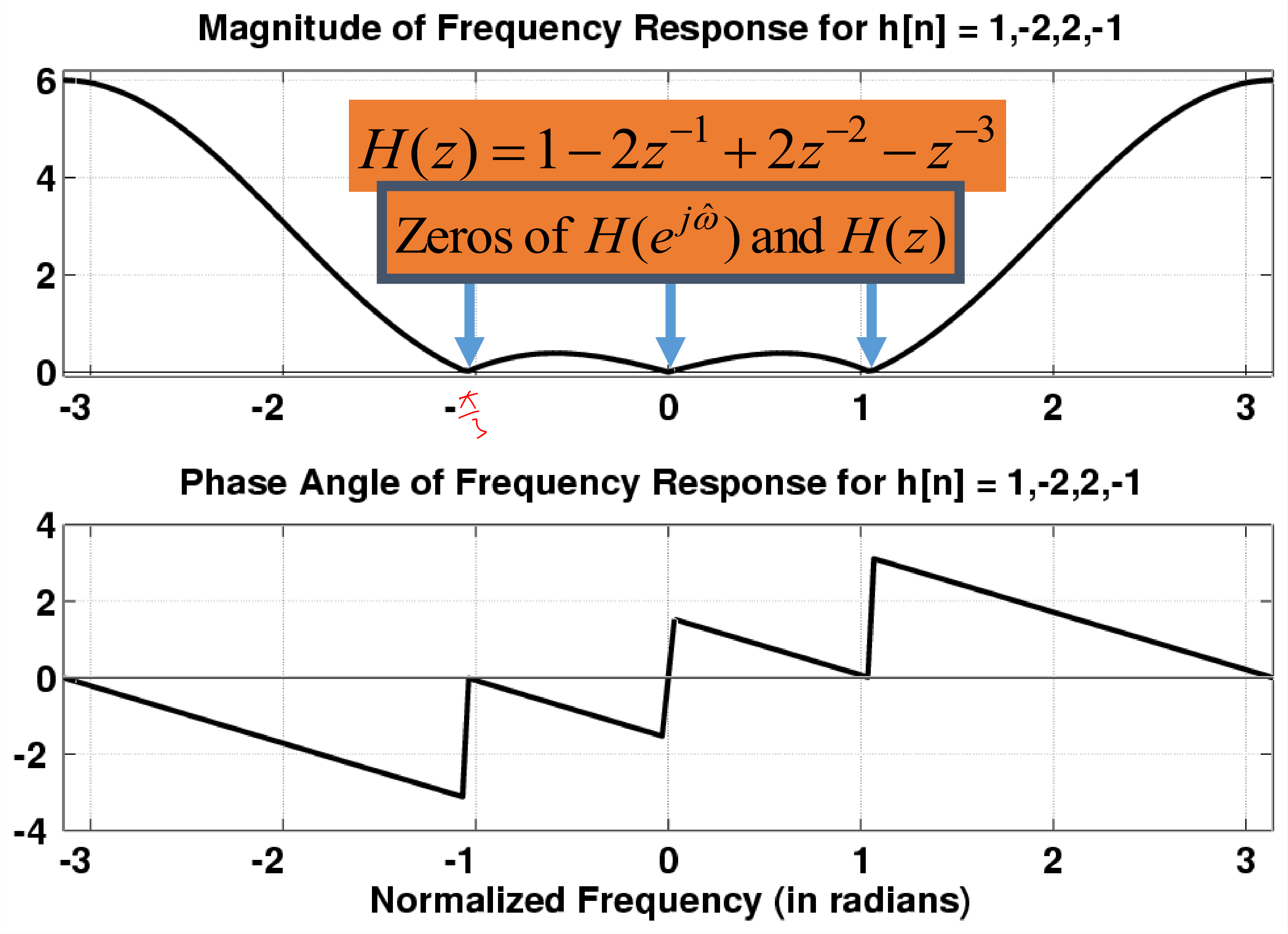
$$y[n] = (1 - 2e^{-j\pi/3} + 2e^{-j2\pi/3} - e^{-j3\pi/3}) \cdot e^{j(\pi/3)n}$$

$$(1 - 2(\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}) + 2(-\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}) - (-1))$$

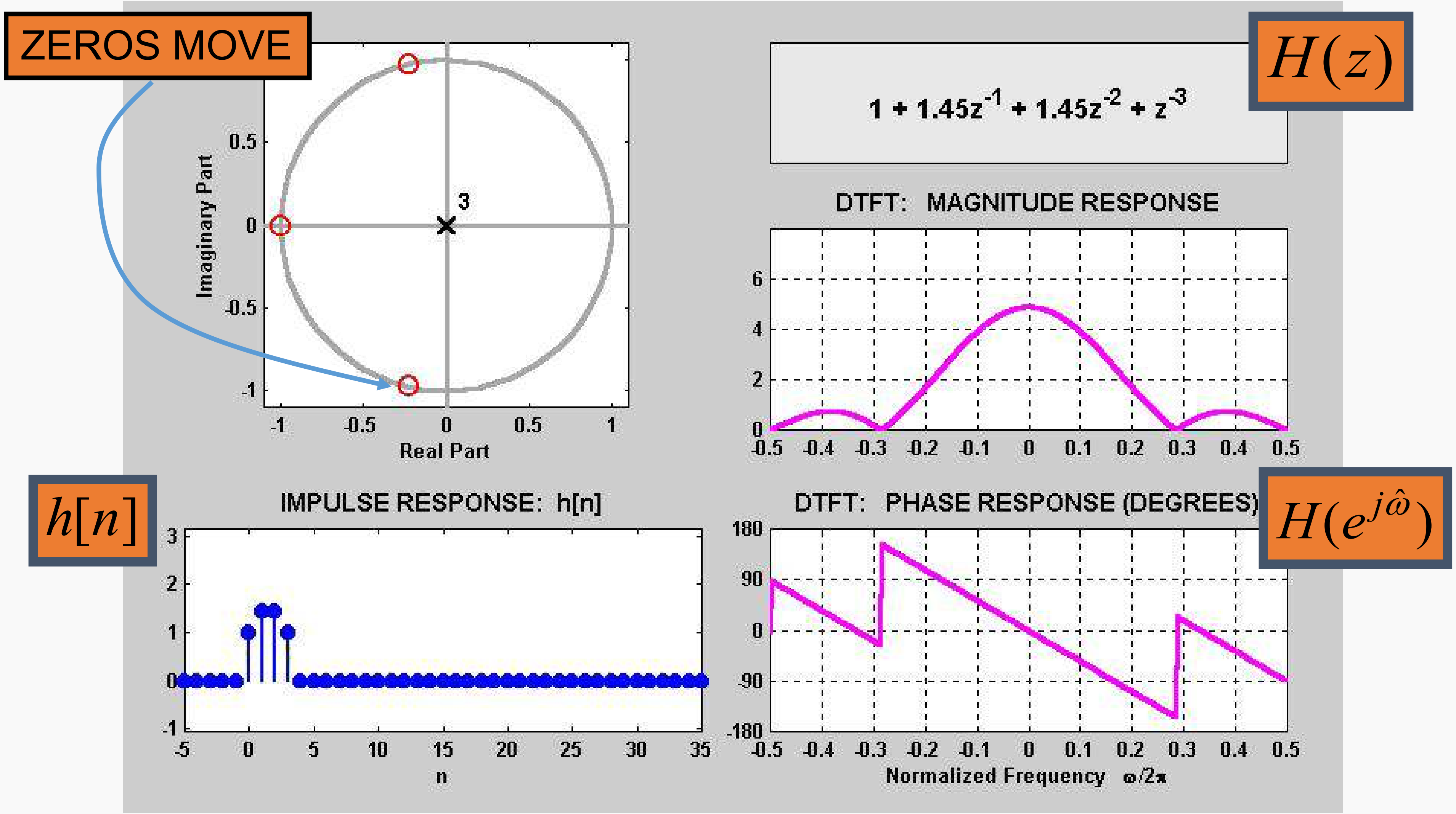
$$y[n] = (1 - 1 + j\sqrt{3} - 1 - j\sqrt{3} + 1) \cdot e^{j(\pi/3)n} = 0$$

© 2003-2016, JH McClellan & RW Schafer

FIR Frequency Response



3 DOMAINS MOVIE: FIR



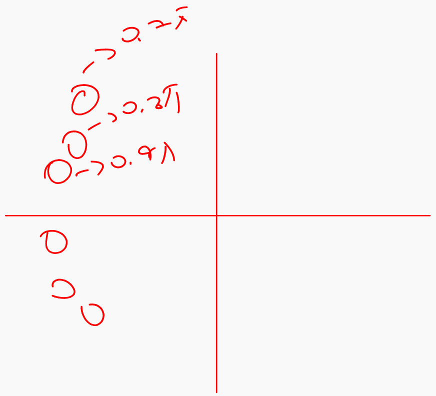
4 MOVIES @ WEBSITE

- http://dspfirst.gatech.edu/chapters/07ztrans/demos/3_domain/index.html
- 3 DOMAINS MOVIES: FIR Filters
  - Two zeros moving around UC and inside
  - Three zeros; one held fixed at $z = -1$
  - Ten zeros; 9 equally spaced around UC; one moving
  - Ten zeros; 8 equally spaced around UC; two moving

© 2003-2016, JH McClellan & RW Schafer

DESIGN PROBLEM

- Example:
  - Design a Lowpass FIR filter (Find b_k)
  - Reject completely 0.7π, 0.8π, and 0.9π
  - Estimate the filter length needed to accomplish this task. How many b_k ?



- Z POLYNOMIALS provide the TOOLS

© 2003-2016, JH McClellan & RW Schafer

NULLING FILTER DESIGN

- PLACE ZEROS to make y[n] = 0

Need 6 ZEROS
where H(z) = 0

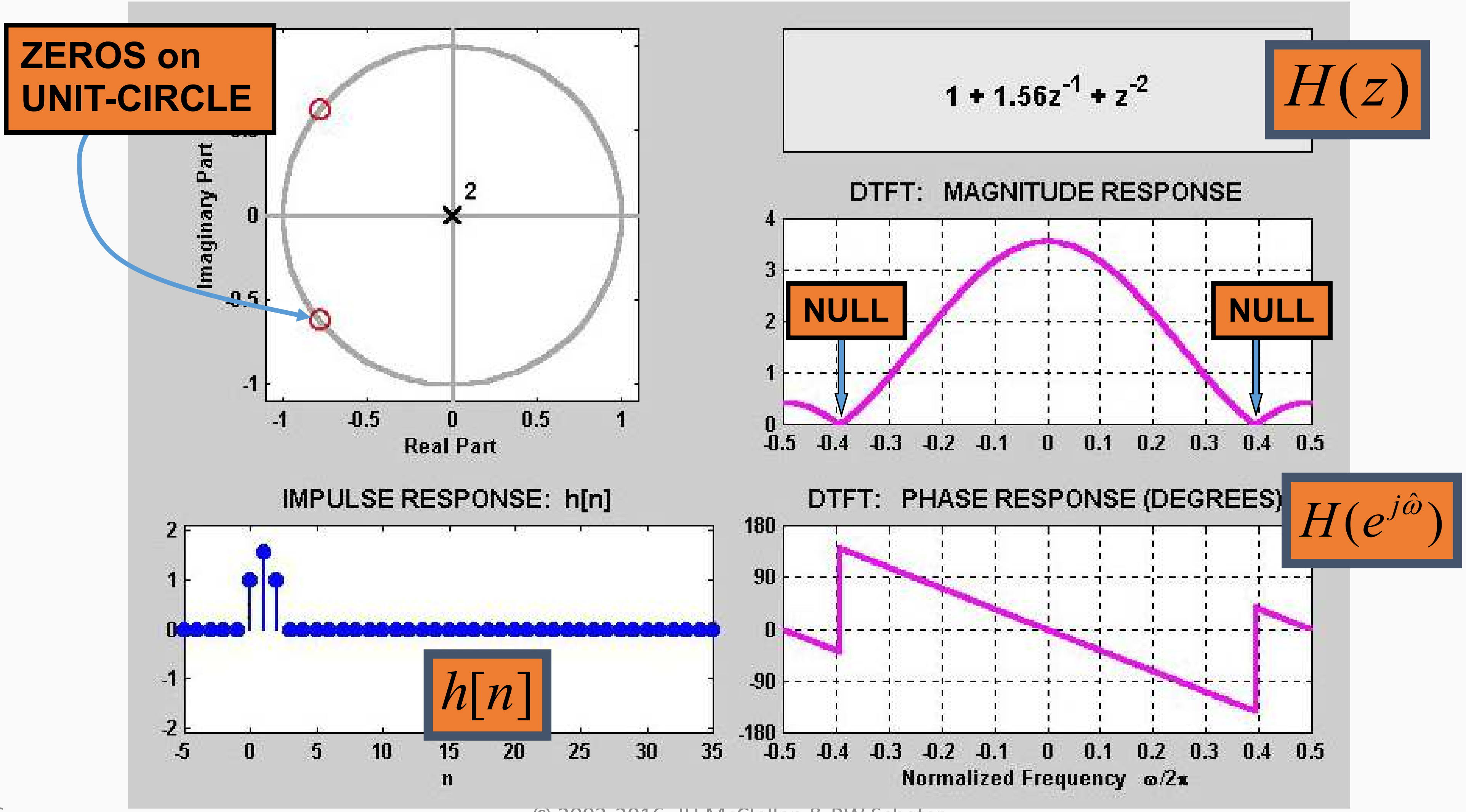
$$H(z_k) = 0, \quad \text{for } z_k = e^{\pm j0.7\pi}, e^{\pm j0.8\pi}, e^{\pm j0.9\pi}$$

- 6th order FIR has 7 filter coefficients

$$x[n] = e^{j0.8\pi n} \quad \Rightarrow \quad y[n] = H(e^{j0.8\pi})e^{j0.8\pi n}$$

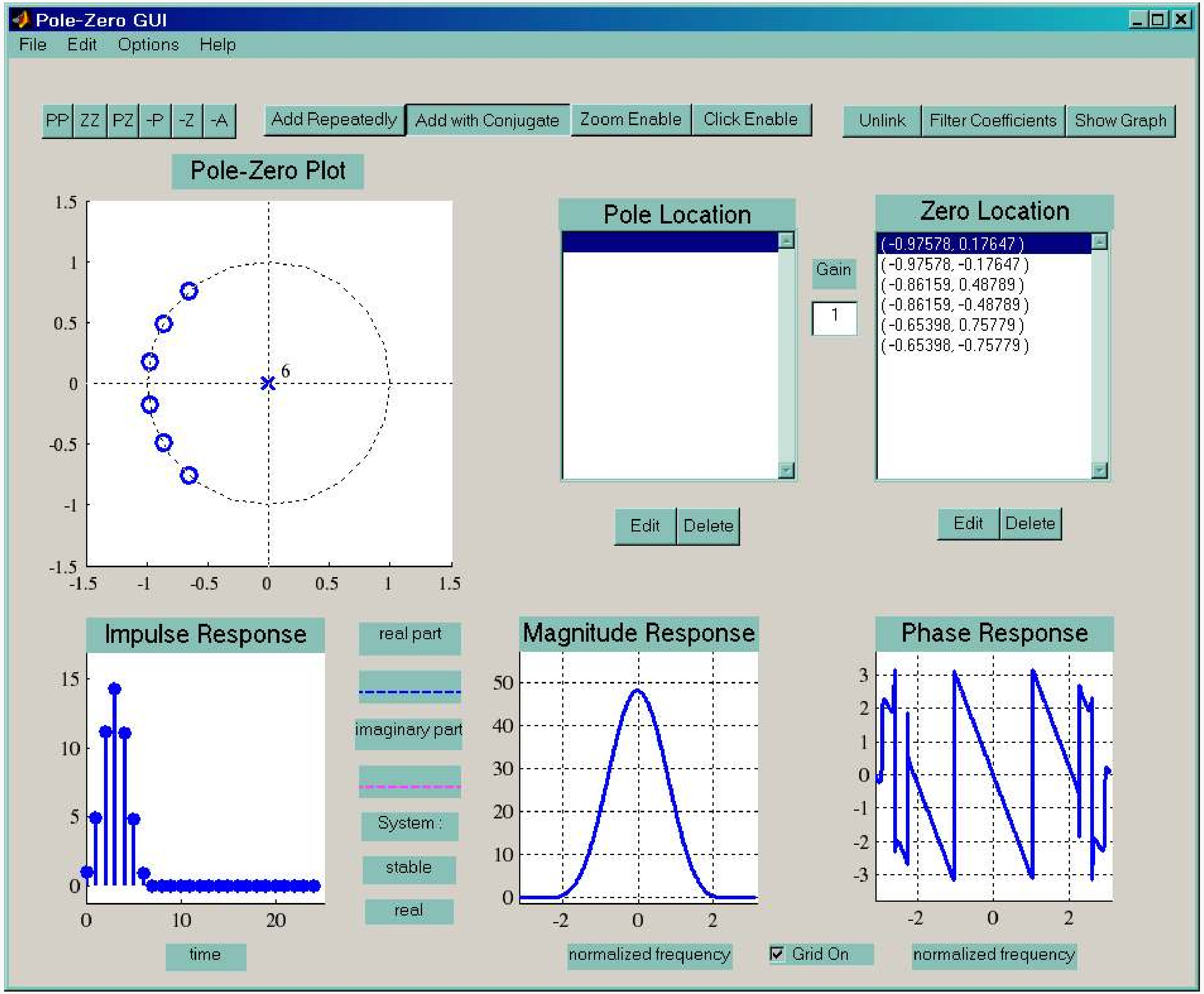
$$H(z) = b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3} + b_4 z^{-4} + b_5 z^{-5} + b_6 z^{-6}$$

3 DOMAINS MOVIE: FIR



© 2003-2016, JH McClellan & RW Schafer

PeZ Demo: Zero Placing



One zero, two zeros, …

We usually want filters with real coefficients

$$\boxed{H(z) = 1 - az^{-1} \quad \Rightarrow \quad H(z) = 0 \;\; @ \;\; z = a}$$

If we want to block sinusoid with $\hat{\omega} = \pm 0.8\pi$

$$H(z_k) = 0 \quad \text{for } z_k = e^{\pm j0.8\pi}$$

$$\Rightarrow H(z) = z^{-2}(z - e^{j0.8\pi})(z - e^{-j0.8\pi})$$

$$= z^{-2}(z^2 - z(e^{j0.8\pi} + e^{-j0.8\pi}) + 1)$$

$$= 1 - 2(\cos 0.8\pi)z^{-1} + z^{-2} = 1 + 1.618z^{-1} + z^{-2}$$

z^{-2} needed for causality

$$h[0] = 1, \quad h[1] = 1.618, \quad h[2] = 1$$

© 2003-2016, JH McClellan & RW Schafer

Block Multiple Frequencies

Want to totally block: $\hat{\omega}_1, \hat{\omega}_2, \cdots, \hat{\omega}_m$

***H*(*z*)** must have zeros at: $z = e^{\pm j\hat{\omega}_1}, e^{\pm j\hat{\omega}_2}, \cdots, e^{\pm j\hat{\omega}_m}$

To block $\hat{\omega} = 0$ or π must have zero at ***z*** =1 or -1

So, the general form becomes:

$$H(z) = (1 - z^{-1})(1 + z^{-1}) \prod_{n=1}^{m} (1 - e^{j\hat{\omega}_n} z^{-1})(1 - e^{-j\hat{\omega}_n} z^{-1})$$

to block DC

to block $f_s/2$

On the other hand: Not much control over other frequencies

© 2003-2016, JH McClellan & RW Schafer

L-pt RUNNING SUM H(z)

$$H(z)=\sum_{k=0}^{L-1} z^{-k}=\frac{1-z^{-L}}{1-z^{-1}}=\frac{z^{L}-1}{z^{L-1}(z-1)}$$

$$\boxed{z^{L}-1=0 \quad \Rightarrow \quad z^{L}=1=e^{j2\pi k}}$$

$$\boxed{z=e^{j(2\pi/L)k} \quad \text{for } k=1,2,\ldots L-1}$$

ZEROS on UNIT CIRCLE

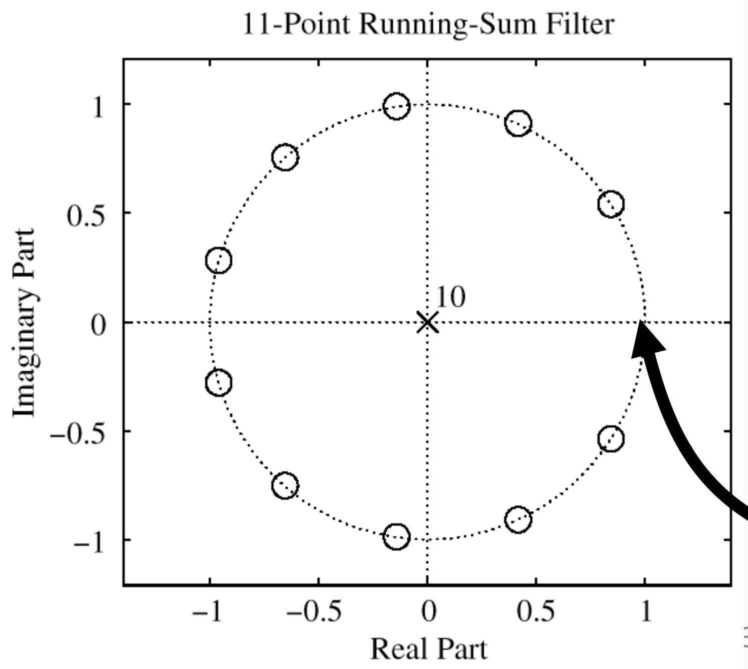
(z-1) in denominator cancels k=0 term

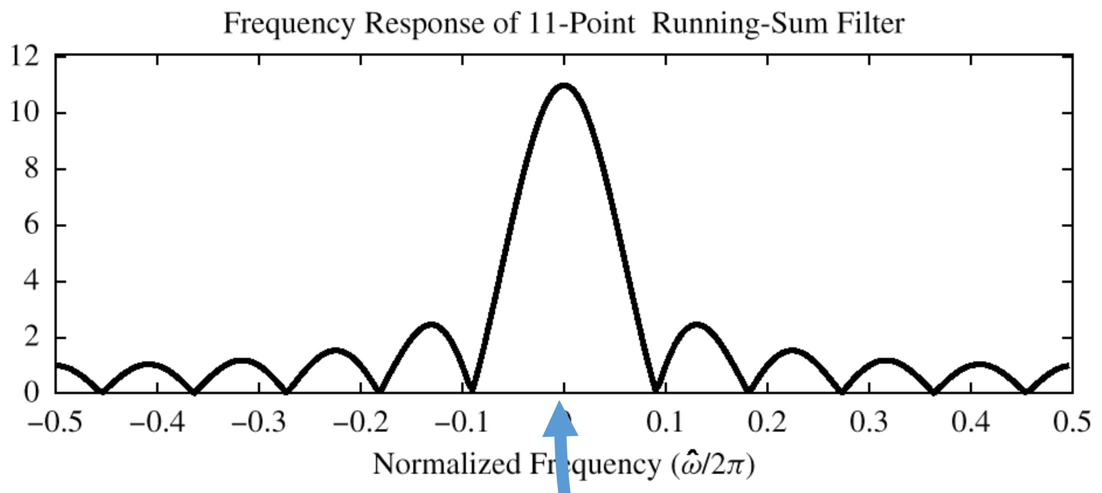
© 2003-2016, JH McClellan & RW Schafer

11-pt RUNNING SUM H(z)

$$H(z) = \sum_{k=0}^{10} z^{-k}$$

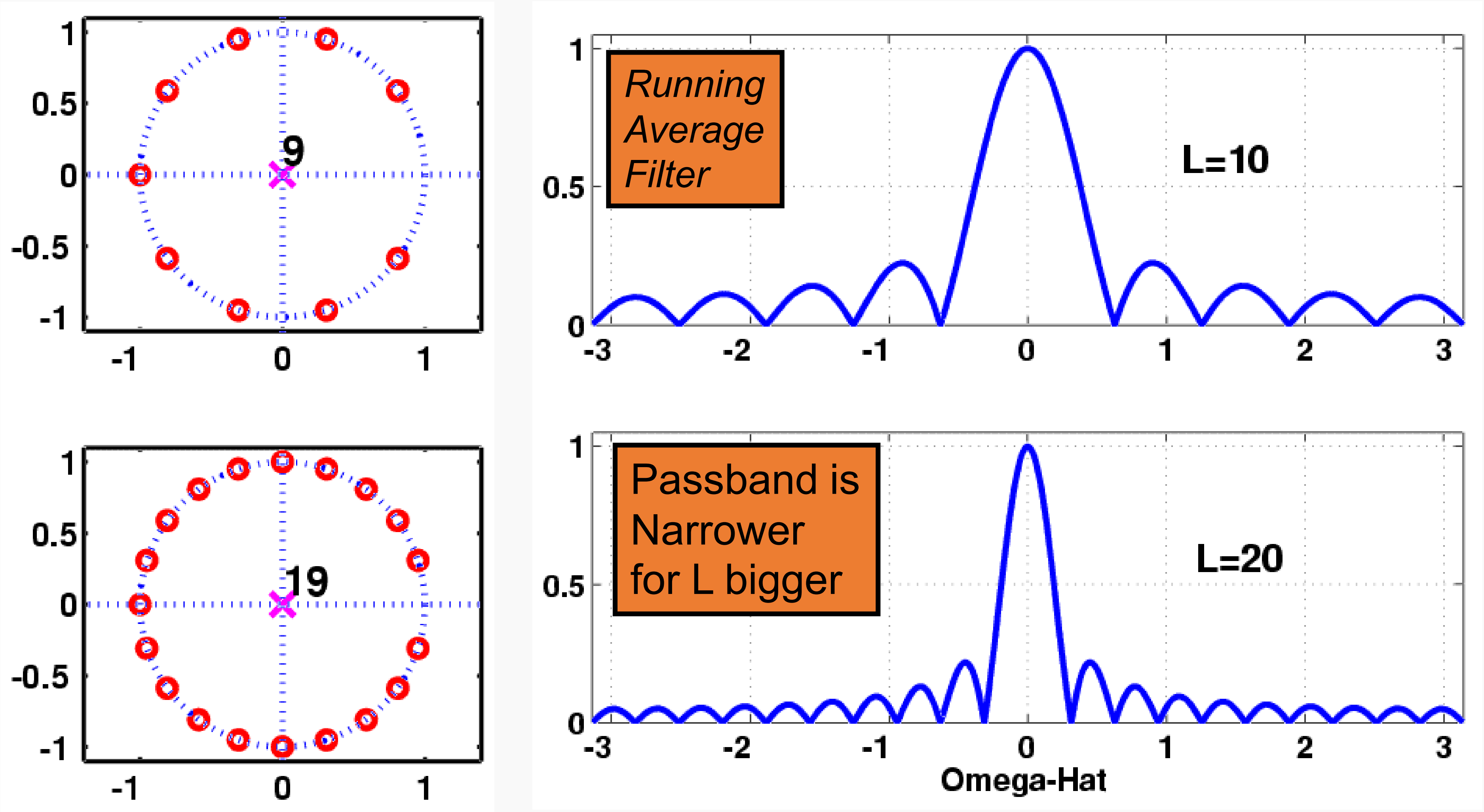
$$H(z) = (1 - e^{j2\pi/11}z^{-1})(1 - e^{j4\pi/11}z^{-1})\cdots(1 - e^{j20\pi/11}z^{-1})$$





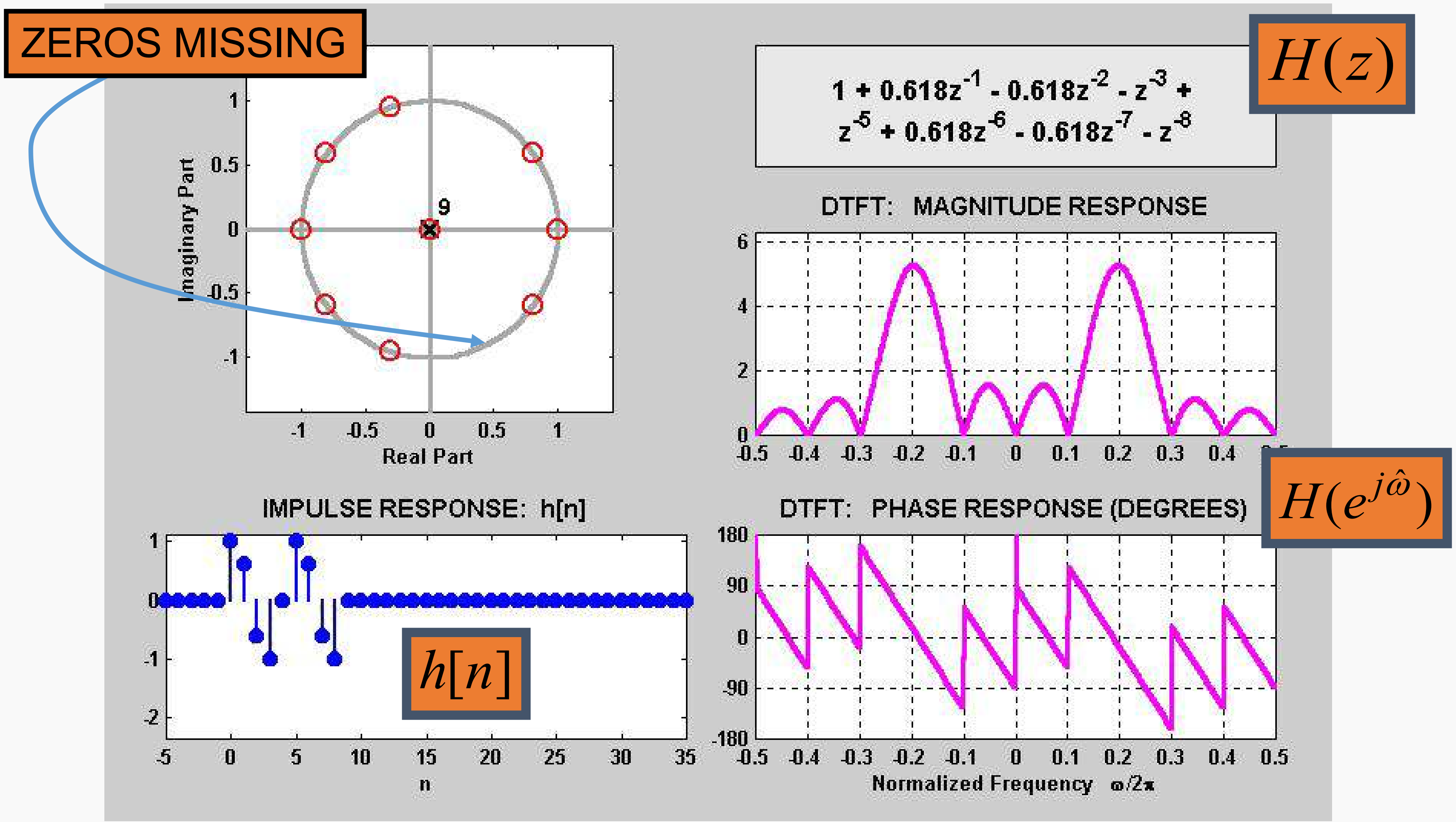
NO zero at z=1

FILTER DESIGN: CHANGE *L*



© 2003-2016, JH McClellan & RW Schafer

3 DOMAINS MOVIE: FIR BPF



Check Website & MATLAB Code

https://dspfirst.gatech.edu/chapters/07ztrans/demos/3_domain/index.html

```
%% load piano sounds
load labtest.mat;
sound(xx,fs);
figure(1); spectrogram(xx);

%% If I want to pass w = 0.2pi
load filter2.mat;
yy = filter( z_values, p_values, xx);
[h,w] = freqz( z_values, p_values, 'whole',120);
plot((w)/pi,abs(h));

sound(yy,fs);
figure(2); spectrogram(yy);
```